Steganography

Iliad Intensive

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Overview

Steganography: Theory

Steganography with LLMs

Steganography: Theory

What is Steganography?

Cryptography

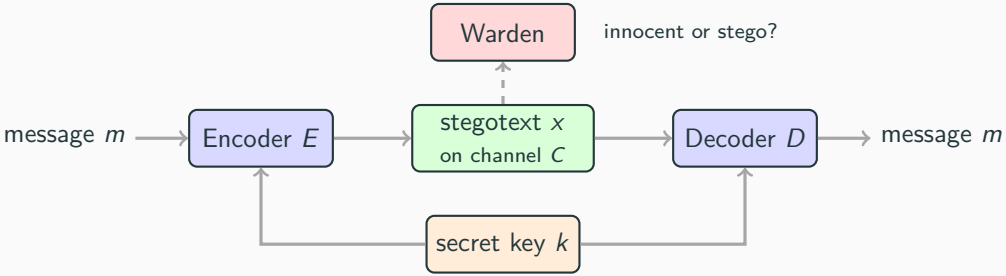
- Hides the **content** of a message
- Outsiders cannot read it
- But everyone sees that a secret is being sent

Steganography

- Hides the **existence** of a message
- The secret sits inside an innocent-looking cover
- An observer sees only ordinary communication

Covers can be images, text, sound files or even network protocols.

The Setup



- Encoder E and decoder D share a secret key k
- A channel C carries the messages
- A distribution $\mathcal{D}$ of innocent messages
- A message space M of possible secrets

- A warden observes the channel
- **Correctness**: decoding succeeds with high probability
- **Security**: the warden's distinguishing advantage is small

Detection as Hypothesis Testing

Innocent channel the normal language model

Covertext the model's response

Hidden message a payload embedded in the response

Adversary sees the response and decides: ordinary or stego?

For a prompt p:

$P = \mathrm{Model}(p)$	innocent
$Q = \mathrm{Steg}_k(p, m)$	stego

The adversary tests

$$H_0 : x \sim P$$
$$H_1 : x \sim Q$$

Perfect Security

The strongest notion is **perfect information-theoretic indistinguishability**:

$$P = Q \quad \iff \quad d_{\mathrm{TV}}(P, Q) = 0$$

with the total variation distance

$$d_{\mathrm{TV}}(P, Q) = \frac{1}{2} \sum_{x} |P(x) - Q(x)|$$

- One sample, equal priors: the best possible adversary succeeds with probability

$$\Pr[\text{correct}] = \frac{1}{2} \left(1 + d_{\mathrm{TV}}(P, Q)\right)$$

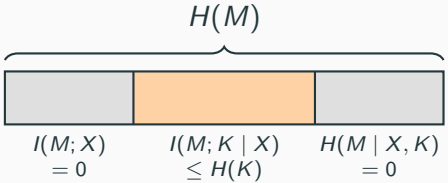
- If $d_{\mathrm{TV}} = 0$, no adversary can do better than guessing, even with unlimited computation

Perfect Security Needs Long Keys

M: message, X: stegotext, K: key

- Perfect security: $I(M;X) = 0$
- Perfect decoding: $H(M \mid X, K) = 0$

$$
\begin{aligned}
H(M) &= H(M \mid X, K) + I(M; X, K) \\
&= \underbrace{H(M \mid X, K)}_{=0} + \underbrace{I(M; X)}_{=0} + I(M; K \mid X) \\
&\leq H(K \mid X) \leq H(K)
\end{aligned}
$$



$$\boxed{H(M) \leq H(K)}$$

Why This Is Impractical

- The key has to grow linearly with the message
- Arbitrarily many hidden messages need an unbounded supply of shared randomness, or a way to refresh it
- This is the one-time pad again: perfect secrecy consumes key material
- Whoever can exchange that much secret randomness could often use that channel directly

Steganography is still not useless: it hides the **existence** of communication, encryption only hides the **content**.

Practical schemes use a short reusable key and settle for **computational security**.

Computational Security

Weaken the requirement:

Perfect

- $P = Q$
- *No one* can distinguish the two distributions

Computational

- $P \approx Q$ for efficient observers
- No *polynomial-time* adversary can distinguish them with non-negligible advantage

An unbounded adversary may still break the scheme.

Pseudorandom Functions

A **PRF** is a family of keyed functions

$$F = \left\{ F_k : \{0,1\}^{\ell_1(\lambda)} \to \{0,1\}^{\ell_2(\lambda)} \;\middle|\; k \in \{0,1\}^{\lambda} \right\}$$

1. F_k is efficiently computable given k
2. For every polynomial-time distinguisher D and a truly random function f:

$$\left| \Pr_k \left[D^{F_k}(1^\lambda) = 1 \right] - \Pr_f \left[D^{f}(1^\lambda) = 1 \right] \right| \leq \mathrm{negl}(\lambda)$$

A short key, but to efficient algorithms it looks like a truly random function.

Assumptions and Attack Cost

Assumption

- One-way functions exist
- This implies that PRFs exist
- $P \neq NP$ alone is too weak for modern cryptography

Security argument

1. Assume secure PRFs exist
2. Replace true randomness by PRF output
3. Efficient adversaries cannot tell the difference
4. So they cannot tell stego from innocent output

Attack cost

- Generic attack: try all keys of length λ

$$O(2^{\lambda})$$

- For each key, check whether the text has the expected statistical structure
- A weak PRF or a flawed construction allows faster attacks
- A good scheme makes exhaustive search the best strategy

Summary: The Trade-off

	Perfect	**Computational**
Guarantee	$P_{\text{stego}} = P_{\text{innocent}}$	no efficient distinguisher
Key	$H(K) \geq H(M)$, fresh	short, reusable
Assumption	none	secure PRFs exist
Unbounded adversary	still secure	breaks it, cost up to 2^{λ}

Steganography with LLMs

PRF + LLM = Undetectable Steganography



- Normal sampling: draw a uniform $u \in [0, 1]$ and pick the token whose interval contains u
- Steganography: replace u by the output of a secret-keyed PRF that depends on the hidden symbol
- The PRF output looks uniform, so the token distribution is unchanged
- Any signal with entropy can be turned into a stego channel this way

"Undetectable Steganography for Language Models" [Zamir, 2024]

Sending and Decoding a Symbol

Sender

- Wants to transmit a symbol $\sigma \in \{0, 1, \leftarrow\}$
- Samples with randomness

$$F_k(r, i, \sigma)$$

- k: secret key, i: token position, r: prefix of this response
- Pushes the same symbol for several consecutive tokens

Receiver

- Knows k, sees the tokens
- Keeps a score for every candidate symbol $0, 1, \leftarrow$
- The score of the correct symbol tends to grow faster
- Once a score crosses a threshold t: record the symbol, reset the scores

"Undetectable Steganography for Language Models" [Zamir, 2024]

Error Correction and Capacity

- **LLM and prompt known**: the receiver can decode perfectly
- **Only the tokenizer known**: the receiver decodes probabilistically and sometimes accepts a wrong symbol
- The payload is encoded with an error-correcting code over $\{0, 1, \leftarrow\}$
- $\leftarrow$ is a **backspace**: it repairs an earlier decoding error
- With an error rate below 0.5 there is net progress towards the correct message

- **Capacity**: longer, higher-entropy responses carry more payload
- **Reuse**: each response starts with a natural high-entropy prefix r that enters all later PRF inputs, so every response gets fresh keyed randomness

"Undetectable Steganography for Language Models" [Zamir, 2024]

From Steganography to Watermarking

A watermark is steganography with a **one-bit payload**: "this text came from our model".

- Same construction: the sampler's randomness comes from a keyed pseudorandom function of the key and the preceding tokens
- Different goal: robust detection, and the adversary wants to *remove* the mark

2022 Scott Aaronson and Hendrik Kirchner propose the Gumbel scheme at OpenAI

2024 SynthID-Text by Google DeepMind, based on tournament sampling

2026 Claude's text watermark, a version of the SynthID-Text approach, announced in August for the EU AI Act transparency rules

My AI Safety Lecture for UT Effective Altruism [Aaronson, 2022]

"Scalable watermarking for identifying large language model outputs" [Dathathri et al., 2024]

The Gumbel Watermark

Generation

- Model gives probabilities p_v
- One pseudorandom number per token v:

$$r = F_k(\text{previous tokens}) \in [0,1]^{|V|}$$

- Output

$$x = \arg\max_v \; r_v^{1/p_v}$$

- Token v wins with probability exactly p_v

Detection

- Needs the key and the text, not the model
- Recompute r for every observed token x_i

$$S = \sum_{i=1}^{n} -\ln\left(1 - r_{x_i}\right)$$

- No watermark: $S \sim \mathrm{Gamma}(n, 1)$, mean n
- Watermark: S is larger, the gap grows with the entropy of the text

Watermarking GPT Outputs [Aaronson and Kirchner, 2023]

Exercise 1: Watermarking

30 minutes



References i

- Aaronson, Scott [2022]. *My AI Safety Lecture for UT Effective Altruism*. URL: https://scottaaronson.blog/?p=6823.
- Aaronson, Scott and Hendrik Kirchner [2023]. *Watermarking GPT Outputs*. Presented at the Simons Institute as "Watermarking of Large Language Models" (talk recording). URL: https://simons.berkeley.edu/talks/scott-aaronson-ut-austin-openai-2023-08-17.
- Dathathri, Sumanth, Abigail See, Sumedh Ghaisas, et al. [2024]. "Scalable watermarking for identifying large language model outputs". In: *Nature* 634. URL: https://www.nature.com/articles/s41586-024-08025-4.
- Zamir, Or [2024]. "Undetectable Steganography for Language Models". In: *Transactions on Machine Learning Research*. URL: https://openreview.net/forum?id=fq6aQoMSHz.